

Integrating artificial intelligence and data envelopment analysis for sustainable efficiency assessment in higher education

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HIGHLIGHTS

- A hybrid framework combining Data Envelopment Analysis and Artificial Intelligence has been developed.
- The model evaluates efficiency and sustainability across higher education courses.
- AI models (Random Forest, Gradient Boosting) accurately predict DEA efficiency scores ($R^2 > 0.90$).
- Sustainability indicators link energy efficiency and digital equity to reduced dropout rates.
- The approach supports data-driven policies aligned with SDG 4 and SDG 12 in higher education.

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ABSTRACT

The rapid digitalization of higher education has exposed inconsistencies between increased investments in technological infrastructure, teaching resources, and energy consumption, and the expected improvements in academic performance. This gap highlights the need for predictive and explainable approaches that can evaluate course-level efficiency while incorporating sustainability principles. In this study, Data Envelopment Analysis (DEA) under the output-oriented BCC formulation was applied to fifteen university courses to estimate their relative efficiency based on multiple inputs, including teaching staff, technological investment, digital resource usage, and energy consumption, and outputs such as passing, retention, completion, satisfaction, and learning outcomes. Only three courses achieved full efficiency ($h = 1.0$), whereas the remaining ones showed efficiency gaps ranging from 5.4% to 20.2% relative to the frontier. Energy consumption and learning outcomes emerged as the most influential determinants of efficiency. Artificial intelligence models were then trained under DEA supervision to predict efficiency scores with high fidelity ($R^2 > 0.90$ for Random Forest and Gradient Boosting), replicating DEA behavior and enabling prospective analysis. Finally, indicators aligned with Sustainable Development Goals (SDGs 4 and 12) demonstrated that equitable digital access management reduces dropout rates. At the same time, optimized energy use enhances sustainability metrics by up to twelve points, reinforcing the relevance of AI-driven learning analytics for efficient and sustainable educational management.

1. Introduction

Efficiency assessment in educational settings has gained increasing relevance in recent years, particularly as artificial intelligence (AI), digitalization, and sustainability pressures reshape the dynamics of teaching and learning. Within these contexts, growing technological and energy investments do not always translate into proportional improvements in

academic outcomes, underscoring the need for analytical frameworks that can identify not only the magnitude of allocated resources but also their relative capacity to generate measurable educational achievements. This challenge becomes especially critical in higher education environments where extensive digital infrastructure coexists with persistent dropout rates, revealing that efficiency cannot be equated with resource abundance but must instead be understood through the lens

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of relative productivity and sustainability. Empirical studies reinforce this view: Visbal-Cadavid et al. (2019) demonstrated that integrating Data Envelopment Analysis (DEA) with neural networks captures heterogeneity across institutions with similar resource endowments. In contrast, Jayashree et al. (2025) demonstrated that managing big data in higher education requires equitable strategies to ensure a sustainable impact. Likewise, Anumolu and Kodi (2025) emphasized that predictive analytics must be directed toward sustainability to prevent ineffective technological investments, and Atta et al. (2023) highlighted that learning analytics can transform digital infrastructure into tangible learning outcomes.

DEA has established itself as a robust and interpretable method for assessing relative efficiency in the presence of multiple heterogeneous inputs and outputs (Camanho et al., 2024; Papaioannou & Podinovski, 2024). Unlike other parametric approaches, DEA constructs an empirical frontier that identifies the most efficient decision-making units (DMUs) and quantifies the distance of the remaining ones from that frontier. In this work, DEA is employed as a structural analytical framework to formally integrate sustainability-oriented dimensions into course-level efficiency assessment, rather than as an end. The output-oriented BCC formulation was applied to capture heterogeneity among university courses by simultaneously considering teaching, technological, and energy resources as inputs, and indicators of academic performance, retention, satisfaction, and learning as outputs. The inclusion of energy variables provides an additional dimension of sustainable resource modeling, aligning the analysis with institutional commitments to responsible digitalization. The heterogeneity detected in the data set confirmed the relevance of the method, as the correlation coefficients between outputs were low (0.05–0.27), ensuring that each indicator provides non-redundant information for estimating the frontier. This empirical robustness further supports DEA's role as a supervisory baseline for the subsequent AI models and reinforces its applicability in educational environments where resource constraints and quality demands coexist (Dixit et al., 2024; Zofio et al., 2024).

The initial DEA assessment revealed substantial heterogeneity among DMUs, indicating that courses differ markedly in their ability to transform teaching, technological, and energy-related resources into educational outcomes. The analysis showed that efficiency is not systematically associated with higher levels of technological investment or staffing, but rather with the capacity to achieve strong learning and completion outcomes while maintaining controlled levels of resource consumption. Energy use emerged as a critical dimension in distinguishing efficiency from less efficient courses, highlighting that sustainability-related factors play a structural role in educational performance. These findings demonstrate the diagnostic capacity of DEA to identify multidimensional efficiency patterns that are not captured by conventional parametric approaches and motivate the subsequent use of AI models to examine efficiency structures under alternative and prospective scenarios.

To extend the static efficiency assessment provided by DEA, artificial intelligence models were employed as a complementary analytical layer. Ensemble-based algorithms and neural networks were used to learn the nonlinear relationships underlying the efficiency structure identified by DEA, enabling the replication and prospective estimation of efficiency scores under alternative resource configurations. Within this framework, AI does not replace the DEA benchmarking logic but rather serves as an interpretable, predictive extension that supports scenario analysis and forward-looking evaluation. Feature relevance analysis further contributes to transparency by identifying the relative influence of academic, technological, and sustainability-related variables on efficiency outcomes, reinforcing the role of AI as a decision-support mechanism rather than as an independent optimization model.

Sustainability was incorporated as a cross-cutting analytical dimension within the hybrid DEA–AI framework, positioning the study within the broader field of AI-driven sustainable learning analytics. Three dimensions were examined: energy efficiency, equity in digital access,

and dropout reduction, each aligned with the Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education) and SDG 12 (Responsible Consumption and Production). The analysis revealed consistent patterns indicating that courses characterized by more efficient energy use and more equitable access to digital resources tend to exhibit higher levels of educational efficiency and student retention. These patterns suggest that social and environmental sustainability are structurally intertwined with academic performance in digitally mediated higher education environments. The complete dataset comprised 500 course-level observations, from which fifteen representative courses were selected for detailed DEA benchmarking and sustainability-oriented analysis, balancing interpretative depth with computational scalability. Together, these results highlight the potential of explainable AI metrics to operationalize the SDGs in higher education by linking resource efficiency, digital inclusion, and learning outcomes within a unified analytical framework.

The integration of DEA with AI in this work is framed as a structured analytical approach that explicitly embeds sustainability-oriented dimensions into course-level efficiency assessment (Eom et al., 2024; Gori et al., 2025). The primary contribution lies in operationalizing energy efficiency, equity in digital access, and dropout reduction as intrinsic components of educational efficiency, rather than as external or *post hoc* evaluation criteria. By using DEA as an interpretable benchmark and AI models as predictive and explainable extensions, the proposed framework enables both retrospective and prospective analyses of sustainable educational performance (Atta et al., 2023; Jayashree et al., 2025). This positioning responds to recent calls in the literature to move beyond resource-centric or purely academic notions of efficiency, and to align educational analytics with broader sustainability and policy objectives, particularly those associated with SDG 4 and SDG 12.

The remainder of this article is structured as follows: Section 2 reviews the literature on DEA, AI, and sustainability in education. Section 3 details the materials and methods employed, including mathematical formulation, experimental setup, and validation procedures. Section 4 presents the results, encompassing both descriptive findings and those derived from the DEA model, AI-based predictions, and sustainability assessment. Section 5 provides a comparative and critical discussion of the results in relation to previous studies and methodological limitations. Finally, Section 6 concludes the paper and outlines directions for future research.

2. Literature review

DEA has established itself as a robust and interpretable method for measuring relative efficiency in educational systems, due to its capacity to integrate multiple heterogeneous inputs and outputs. Recent applications across different academic levels have demonstrated its practical relevance for identifying efficient institutions and benchmarking performance in contexts characterized by increasing digitalization and resource constraints (Gori et al., 2025; Mudawali et al., 2025). In higher education, DEA has been used to evaluate teaching performance, institutional productivity, and resource allocation, providing decision-makers with quantitative evidence to support improvement strategies.

To address some of the limitations of traditional DEA models, particularly their static nature and sensitivity to data variability, several studies have explored integrating DEA with artificial intelligence techniques. Neural networks, deep learning models, and other machine learning approaches have been applied to enhance the discriminatory power and predictive capacity of DEA in educational environments (Nazir et al., 2025; Sun et al., 2023). In parallel, AI-driven analytics have been widely adopted in learning management systems to optimize assessment processes, support academic advising, and personalize learning experiences (Dias et al., 2025; Eom et al., 2024; Zhang, 2024). These contributions show that hybrid DEA–AI approaches can extend efficiency analysis toward prospective and adaptive educational analytics.

Alongside these methodological developments, a growing body of literature has emphasized the role of sustainability in higher education, particularly in relation to digital transformation, equity, and responsible resource use. Studies such as those by Anumolu and Kodi (2025) and Jayashree et al. (2025) demonstrate how predictive analytics and big data can support sustainable educational practices by reducing dropout rates, optimizing institutional resources, and promoting inclusive access to digital learning. Other works highlight the contribution of AI and data science to educational resilience and social sustainability, especially during periods of systemic disruption (Aguilar-Esteva et al., 2023). From an institutional perspective, learning analytics has also been shown to contribute to both pedagogical and organizational sustainability, reinforcing alignment with the SDGs (Atta et al., 2023; Bahari et al., 2023).

In terms of practical applications, research across diverse educational contexts has validated hybrid DEA–AI models as valuable instruments for predicting academic achievement and assessing relative efficiency. Outcome prediction through AI has been shown to promote sustainable educational practices (Albahli, 2025). At the same time, Aeni et al. (2024) linked the use of artificial intelligence to innovation and growth in education within a sustainable development framework. These contributions reinforce the potential of hybrid models to support both institutional efficiency and educational sustainability in applied settings.

Despite these advances, the intersection between DEA-based efficiency assessment and sustainability-oriented educational analytics remains only partially explored. While existing DEA–AI studies demonstrate improvements in predictive accuracy and robustness, sustainability is often treated as a contextual consideration or as an external evaluation criterion rather than as an intrinsic component of efficiency assessment. Moreover, practical challenges related to data heterogeneity, governance, and scalability continue to limit the adoption of AI-driven efficiency models in educational systems (Esakkiammal & Kasturi, 2024; Gouveia et al., 2025; Visbal-Cadavid et al., 2019). These limitations highlight the need for analytical frameworks that explicitly integrate sustainability-oriented dimensions, such as energy efficiency, equity in digital access, and dropout reduction, within DEA-based educational efficiency analysis, particularly in alignment with SDG 4 and SDG 12 (Gallego et al., 2024; Martínez-Usarralde & Lloret-Catalá, 2020).

3. Materials and methods

3.1. Dataset

The experimental framework was developed following a hybrid data acquisition strategy that combined open educational repositories with complementary records obtained from controlled experiments in learning management environments. This approach ensured both external validity and contextual representativeness, capturing real patterns of digital platform usage while incorporating sustainability-oriented variables not commonly found in public datasets.

The primary source was the LearnPlatform Educational Technology Dataset (openCPSR), a well-established repository documenting the adoption and utilization of educational technologies across institutions in the United States (LearnPlatform Inc. & Styers, 2021). This dataset provided detailed indicators of student interaction frequency, technological diversity, and participation rates in diverse institutional settings. To broaden coverage, anonymized data from openly available massive open online courses (MOOCs) were integrated, adding large-scale indicators of access, engagement, and completion in virtual learning environments.

To incorporate sustainability dimensions, absent in open repositories, complementary data were generated from controlled trials conducted in university courses managed via the Moodle platform. These records included estimated energy consumption of the digital infrastructure, results from standardized student satisfaction surveys, and inter-semester retention rates. Their design followed empirical distributions reported in previous studies and institutional benchmarks,

ensuring both statistical coherence and consistency between resource allocation and educational performance.

The final dataset comprised 500 course-level observations, each corresponding to a semester-long course. Approximately 70% of the data originated from open repositories, while the remaining 30% stemmed from complementary Moodle-based experiments. This proportion maintained a robust empirical foundation while enriching the analysis with sustainability-oriented indicators essential for evaluating efficiency from a broader socio-technical perspective.

The recorded variables covered both input and output dimensions. Inputs represented academic and technological structures, including the number of assigned instructors, semester-level investment in digital resources, the number of active technological tools, average weekly student connection time, and estimated server energy consumption. Outputs captured the educational effectiveness of each course, comprising the pass rate, inter-semester retention, student satisfaction (on a 1–5 Likert scale), course completion rate, and a composite learning indicator derived from the weighted grade point average.

The selection of these variables aimed to establish a quantitative framework that reflects the balance between institutional investment and pedagogical outcomes. Input captured the institutional and technological effort sustaining the learning process, while output represented measurable academic and social achievements. Integrating both dimensions within the DEA–AI framework enabled a comparative assessment of course-level efficiency and, through predictive modeling, offered insights into how digital transformation and sustainable resource management converge to enhance learning performance and environmental responsibility.

From the complete set of 500 course-level observations, a subset of fifteen courses was selected for detailed DEA benchmarking and subsequent AI-based modeling, to ensure representativeness across key structural dimensions relevant to efficiency assessment. The chosen courses span different disciplinary orientations, including theoretically oriented subjects and practice-intensive courses, and diverse delivery modalities, including fully virtual and blended formats. In addition, the selection intentionally covered a wide range of technological intensity and energy consumption profiles, ensuring that courses with low, medium, and high digital resource demands were included.

The selection process prioritized courses with complete and consistent records across all input and output variables for the entire semester, thereby avoiding distortions in the construction of the efficiency frontier caused by missing or unstable data. This strategy balances computational tractability with interpretative depth, allowing the DEA model to capture meaningful relative efficiency differences while preserving heterogeneity in course structure and resource use.

It is important to note that DEA efficiency scores are inherently relative to the sampled units. Accordingly, the differences observed in efficiency may reflect both resource allocation practices and structural characteristics of the course. The use of an output-oriented BCC model mitigates scale-related effects by allowing variable returns to scale, thereby reducing bias associated with course size or intensity. Consequently, efficiency results should be interpreted as comparative indicators of how effectively courses transform academic, technological, and energy-related inputs into educational outcomes, rather than as absolute measures detached from structural context.

3.1.1. Data preprocessing

The dataset underwent a multi-stage cleaning and transformation procedure to ensure statistical coherence and suitability for both DEA and AI-based analyses. In the initial stage, the completeness and consistency of all records were verified. Outliers were identified by analyzing deviations from the mean and computing 95% confidence intervals, which facilitated the detection of inconsistent or extreme observations. Missing values in continuous variables were imputed using arithmetic means, while categorical variables were completed using mode imputation. When the proportion of missing data exceeded

5%, the corresponding records were removed under controlled deletion to prevent bias. As an additional validation, lightweight machine learning techniques such as the k -nearest neighbors (kNN) imputer were tested for robustness. However, traditional statistical imputation methods were retained due to their higher stability in medium-sized datasets (Chakrabarti et al., 2023).

To homogenize variable scales and prevent disproportionate influence of different magnitudes in the efficiency calculation, all continuous variables were normalized using min–max scaling to the range [0,1], computed as $(x_i - x_{\min}) / (x_{\max} - x_{\min})$, where $x_{\min}$ and $x_{\max}$ denote the minimum and maximum observed values. This transformation ensured comparability among heterogeneous variables such as technological investment, connection hours, and energy consumption.

As a sensitivity check, standardization based on z-scores was also evaluated, computed as $(x_i - \mu) / \sigma$, where μ and σ represent the mean and standard deviation of the variable, respectively. Because DEA requires preserving proportional relationships between inputs and outputs, min–max normalization was retained as the primary preprocessing method, while z-score standardization was used exclusively for robustness validation.

Data preprocessing also involved transforming log files extracted from learning management systems (LMS). Logs from the Moodle platform provided granular information on access frequency, connection time, and participation in digital activities. These records were temporally aggregated to generate course-level weekly indicators, such as mean weekly accesses and average connection times, ensuring comparability across courses and semesters. Similar aggregation procedures were applied to participation in assessments and digital activities, yielding harmonized metrics aligned with the input definitions of the DEA model.

After preprocessing, the DMUs were defined. Each semester-long course was treated as a DMU characterized by a vector of inputs and outputs (Rashid et al., 2024), mathematically represented as:

$$DMU_j = (x_{1j}, x_{2j}, \dots, x_{mj}; y_{1j}, y_{2j}, \dots, y_{sj}) \quad (1)$$

where x_{ij} represents the m input variables and y_{rj} the s output variables of unit j . The semester-level course was selected as the unit of analysis to achieve an intermediate level of aggregation, balancing resource investment and learning outcomes. Analysis at the student level would have introduced excessive heterogeneity, while institutional-level aggregation would have obscured intra-program variability. Course-level DMUs, therefore, provided an optimal resolution for capturing relative efficiency variations across academic programs while preserving methodological consistency.

3.1.2. Data Envelopment Analysis (DEA) model

The efficiency assessment was conducted using DEA, a nonparametric linear programming technique widely applied to evaluate the relative efficiency of DMUs that transform multiple heterogeneous inputs into various outputs. In this study, each semester-long course was modeled as a DMU characterized by academic and technological inputs and by outputs reflecting both academic performance and sustainability-oriented achievements.

An output-oriented model was implemented with the primary objective of maximizing educational outcomes, including passing rates, retention, satisfaction, and completion, while effectively utilizing available resources. The relative efficiency of a given DMU o was defined as the ratio of a weighted sum of outputs to a weighted sum of inputs:

$$\max_{u,v} h_o = \frac{\sum_r 1^s u_r y_{ro}}{\sum_i 1^m v_i x_{io}} \quad (2)$$

subject to the following constraints:

$$\frac{\sum_r 1^s u_r y_{rj}}{\sum_i 1^m v_i x_{ij}} \leq 1, \quad j = 1, \dots, n \quad (3)$$

$$u_r, v_i \geq \epsilon > 0, \quad \forall r, i \quad (4)$$

where y_{rj} denotes the value of output r in DMU j , x_{ij} represents input i , and u_r, v_i are the weights endogenously determined by the model.

Two classical variants of DEA were tested. The first, the Charnes, Cooper, and Rhodes (CCR) formulation, assumes constant returns to scale and compares all DMUs under strict proportionality between inputs and outputs (Rashid et al., 2024). The second, the Banker, Charnes, and Cooper (BCC) model, introduces variable returns to scale, allowing efficiency to be evaluated under both expansion and contraction of resources (Sharafeddin et al., 2024). Since educational processes rarely exhibit strict proportionality—an increase in digital investment or teaching staff does not necessarily yield equivalent academic improvements—the BCC model was adopted as the most suitable formulation for this analysis.

Efficiency scores were obtained by solving the optimization problem for each DMU. Units with $h_o = 1$ were classified as efficient, while those with $h_o < 1$ were considered inefficient and compared with convex combinations of units that define the empirical efficiency frontier. This mechanism enabled both the identification of benchmark courses and the quantification of performance gaps among less efficient ones.

Adapting DEA to a sustainable educational framework required a deliberate definition of variables. Inputs included the number of instructors, investment in educational technology, the variety of digital resources used, average weekly student connection time, and the estimated energy consumption associated with digital infrastructure. Outputs encompassed passing rates, student retention, survey-based satisfaction, course completion rates, and a learning index computed from weighted grade point averages. Consequently, the efficiency scores captured not only academic productivity but also each course's capacity to transform limited technological and energy resources into outcomes aligned with the principles of educational sustainability.

Fig. 1 illustrates the conceptual structure of the model. Inputs feed the DMU for each course, which generates outputs in both academic and digital performance. The comparative evaluation across all units determines the efficient frontier which serves as a reference for sustainable educational management.

3.2. Integration with artificial intelligence

3.2.1. Applied artificial intelligence techniques

AI was integrated to extend the predictive and analytical capabilities of the DEA framework. This integration enabled not only the replication of efficiency scores but also the identification of underlying patterns that govern academic and sustainability-related performance. Two complementary paradigms were implemented: tree-based machine learning models and deep neural architectures.

Within the first paradigm, a Random Forest Regressor was employed due to its robustness in modeling nonlinear relationships between inputs and outputs, as well as its resistance to overfitting and sensitivity to noise. Within seconds, a deep feedforward neural network was trained with two hidden layers, each comprising 128 and 64 neurons, respectively. The architecture used ReLU activation functions in the hidden layers and a linear activation in the output layer. Network training was conducted using error backpropagation optimized by the Adam algorithm, with an early-stopping criterion governed by the mean squared error (MSE).

The dataset was partitioned into 70% for training and 30% for validation, and a five-fold cross-validation strategy was applied to minimize variance and ensure generalizability. Evaluation relied on the coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE), providing objective grounds for comparing model performance. Empirically, the Random Forest model exhibited higher stability with medium-sized datasets, while the neural network achieved superior results when incorporating unstructured features derived from natural language processing. The predictive framework was formally expressed as:

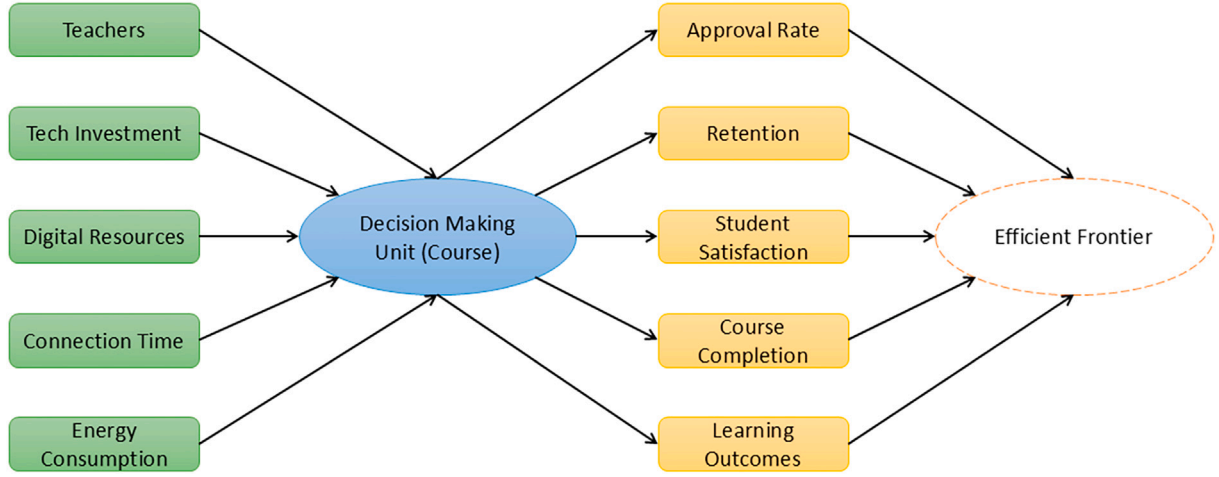


Fig. 1. DEA model adapted to educational platforms, showing inputs, the decision-making unit (course), outputs, and the efficient frontier within a sustainability-oriented context.

$$\hat{E} = f_{\theta}(Z) \quad (5)$$

where $Z = X, Y, F$ denotes the extended feature vector—including DEA inputs (X), outputs (Y), and additional features (F) derived from unstructured data— θ represents the learned model parameters, and $\hat{E}$ is the predicted efficiency value.

Beyond predictive performance, model interpretability was prioritized as a fundamental aspect in educational and sustainability-oriented contexts. For the Random Forest model, feature importance analysis was conducted to identify variables with the most significant influence on predicted efficiency. In the case of neural networks, post-hoc interpretability methods based on SHAP values were applied to estimate the marginal contribution of each variable to the model output. This focus on explainability was essential, as AI integration in educational management must ensure not only predictive accuracy but also transparency and ethical accountability, providing interpretable insights for sustainable decision-making and resource optimization.

During model training, critical hyperparameters were systematically tuned to ensure stability and generalization capacity. For the Random Forest model, 500 trees were defined, with a maximum depth of 20 and a minimum of 5 samples per terminal node, thereby achieving a balance between precision and control over overfitting. For the deep neural network, training used a batch size of 32, an initial learning rate of 0.001, L2 regularization with $\lambda = 0.0001$, and a dropout rate of 20% in the first hidden layer. These configurations were selected through preliminary cross-validation and maintained constant in subsequent experimental runs, ensuring a transparent, reproducible, and interpretable framework for predicting DEA-derived efficiency.

3.2.2. DEA–AI complementarity

The integration of DEA with AI was designed to achieve three interconnected objectives that extend the analytical capacity of classical efficiency evaluation. First, the hybrid framework enables the prediction of future efficiency by training AI models on historical DEA results, allowing anticipation of course performance under specific resource configurations. Second, it enhances the discriminatory power among DMUs, addressing a well-known limitation of traditional DEA models that often assign full efficiency to multiple units. Third, it incorporates unstructured data, such as participation metrics from virtual forums and qualitative insights from open-ended satisfaction survey responses, through natural language processing (NLP) pipelines that generate additional structured indicators integrated into the AI model inputs.

The complete integration workflow is summarized in the pseudocode presented below, which outlines preprocessing, efficiency estimation, model training, and prediction stages:

Algorithm 1 DEA_AI_Integration.

Require: Dataset $D = \{X, Y, U\}$ with inputs X , outputs Y , and unstructured data U

Ensure: Ranked DMUs, predicted efficiencies, and improvement guidelines

- 1: **Step 1: Preprocess D**
 - 2: Handle missing values (imputation/elimination)
 - 3: Normalize inputs and outputs (min–max scaling)
 - 4: Extract features from logs (average weekly activity, connection time)
 - 5: Process unstructured data U using NLP $\rightarrow$ Features F
 - 6: Construct extended dataset $D' = \{X, Y, F\}$
 - 7: **Step 2: Apply DEA to D'**
 - 8: Define DMUs as courses
 - 9: Compute efficiency scores E using BCC model
 - 10: **Step 3: Train AI model f_{θ}**
 - 11: Input: $\{X, Y, F\}$, Target: E
 - 12: Select algorithm: Random Forest / Neural Network
 - 13: Optimize parameters θ
 - 14: **Step 4: Predict efficiency for new data**
 - 15: Input: $\{X_{new}, Y_{new}, F_{new}\}$
 - 16: Output: $\hat{E} = f_{\theta}(X_{new}, Y_{new}, F_{new})$
 - 17: **Step 5: Integrate DEA and AI results**
 - 18: Compare $\hat{E}$ with DEA efficiency
 - 19: Identify discrepancies and refine rankings
 - 20: Generate recommendations for resource allocation
-

This hybrid procedure consolidates the robust benchmarking capacity of DEA with the adaptive and predictive flexibility of AI, enabling comprehensive evaluation and forecasting of academic efficiency. DEA establishes a reference frontier of optimal performance, while AI extends its analytical reach by learning complex nonlinear patterns and incorporating features from unstructured educational data. The resulting framework enables not only a comparative assessment of courses but also proactive scenario modeling and sustainable decision-making, grounded in data transparency and interpretability.

3.3. Sustainability assessment

The sustainability assessment was designed as a multidimensional framework integrating quantitative and qualitative indicators to capture not only the academic efficiency of educational platforms but also their

contribution to responsible resource management and equitable access. All indicators were explicitly aligned with the SDGs, particularly SDG 4, which promotes inclusive and equitable quality education, and SDG 12, which advocates for responsible consumption and production (Obaideen et al., 2022).

Energy efficiency was identified as a fundamental component of sustainability. The energy consumption of servers and associated digital infrastructure for each course was estimated using workload-to-kilowatt-hour (kWh) conversion models. For each DMU, the following relationship was employed:

$$EC_j = \alpha \cdot T_j + \beta \cdot U_j \quad (6)$$

where EC_j denotes the estimated energy consumption of DMU j , T_j represents the total accumulated student connection time, U_j reflects the utilization level of digital resources, and α, β are conversion coefficients calibrated using benchmark energy values for educational computing environments. This indicator enabled the quantification of the energy intensity of each course and the assessment of its influence on sustainable efficiency scores within the DEA-AI framework.

The second sustainability dimension addressed equity of access. To operationalize this concept, dispersion and coverage metrics of student participation were computed. Specifically, an equity index was defined as:

$$Eq_j = 1 - \frac{\sigma_{p_j}}{\mu_{p_j}} \quad (7)$$

where σ_{p_j} is the standard deviation of student participation in course j and μ_{p_j} is the mean participation rate. Values approaching 1 indicate equitable access and balanced utilization of resources, whereas lower values reveal disparities in student participation. This metric aligns directly with the inclusion and accessibility dimensions of SDG 4.

The third criterion focused on student retention as a proxy for dropout reduction, measured as the proportion of enrolled students who completed each course. The retention rate was defined as:

$$R_j = \frac{N_{final,j}}{N_{inicio,j}} \quad (8)$$

where $N_{final,j}$ is the number of students who completed course j , and $N_{inicio,j}$ is the number initially enrolled. This indicator was incorporated as an output variable in the DEA model, representing each course's capacity to sustain complete educational trajectories and minimize attrition.

In addition to the quantitative indicators, qualitative measures derived from student satisfaction surveys and instructor interviews were incorporated. These qualitative responses were coded and transformed into ordinal variables, complementing the quantitative analysis. This integration enabled the evaluation of less tangible but essential dimensions of sustainability, such as perceived digital inclusion, workload fairness, and the adequacy of available technological resources.

These indicators enabled a comprehensive evaluation of sustainability at the course level. Quantitative metrics provided objective measurements of energy consumption, equity, and retention, while qualitative indicators contributed contextual insight into user experience and perceived fairness in digital learning environments. The articulation of these dimensions within the SDG 4 and SDG 12 framework ensured that the efficiency scores estimated through DEA and enhanced via AI techniques were interpreted not only as indicators of academic performance but also as reflections of institutional responsibility, environmental stewardship, and social equity in higher education.

3.4. Model validation

Validation of the integrated DEA-AI framework was conducted using a dual strategy, combining predictive evaluation of the AI component and robustness testing of the DEA-generated efficiency scores. For

the AI validation, the dataset D was divided into training and testing subsets, D_{train} and D_{test} , in a 70:30 ratio. Additionally, a k -fold cross-validation scheme was applied, where D was partitioned into subsets $D_1, D_2, \dots, D_k$. At each iteration, the model was trained using $D \setminus D_i$ and validated on D_i . The average cross-validation error was computed as:

$$CV_{error} = \frac{1}{k} \sum_{i=1}^k L(f_{\theta_i}(D_i), y_i) \quad (9)$$

where L represents the loss function (mean squared error), and f_{θ_i} corresponds to the model trained in iteration i .

Model performance was quantified through standard regression metrics. The coefficient of determination was calculated as:

$$R^2 = 1 - \frac{\sum_{j=1}^n (y_j - \hat{y}_j)^2}{\sum_{j=1}^n (y_j - \bar{y})^2} \quad (10)$$

The mean absolute error (MAE) and root mean square error (RMSE) were computed as:

$$MAE = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|, \quad RMSE = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2} \quad (11)$$

In classification scenarios distinguishing between efficient and inefficient units, precision (P), recall (R), and the F1 score are used:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R} \quad (12)$$

where TP , FP , and FN represent true positives, false positives, and false negatives, respectively.

The DEA model was validated through sensitivity analyses, where input and output sets were systematically varied for each DMU. Robustness was evaluated by computing the standard deviation of efficiency scores across alternative configurations:

$$\sigma_{DEA}(h) = \sqrt{\frac{1}{M} \sum_{m=1}^M (h_m - \bar{h})^2} \quad (13)$$

where h_m denotes the efficiency score obtained under configuration m , and $\bar{h}$ is the average efficiency. Low values of $\sigma_{DEA}(h)$ indicate strong model stability and robustness to changes in the selected variables.

To ensure consistency between the two analytical components, the efficiency scores estimated by DEA (h_j) were compared with those predicted by the AI models ($\hat{h}_j$) for each DMU j . The mean absolute deviation between both sets of scores was calculated as:

$$Error_{DEA-AI} = \frac{1}{n} \sum_{j=1}^n |h_j - \hat{h}_j| \quad (14)$$

This comparison quantified the degree of alignment between DEA-derived and AI-predicted efficiencies, confirming the methodological coherence of the integrated framework. The resulting low error values demonstrated that the hybrid DEA-AI model produced consistent, accurate, and stable outputs suitable for supporting decision-making in sustainable educational environments.

Fig. 2 illustrates the complete validation workflow, summarizing the methodological process from initial data partitioning and cross-validation to performance evaluation, DEA robustness analysis, and final consistency comparison between efficiency scores.

3.5. Experimental tools and environment

The experimental procedures were executed in an open-source and high-performance computing environment specifically configured for the hybrid DEA-AI framework. Python 3.10 served as the primary programming language due to its versatility and extensive ecosystem of

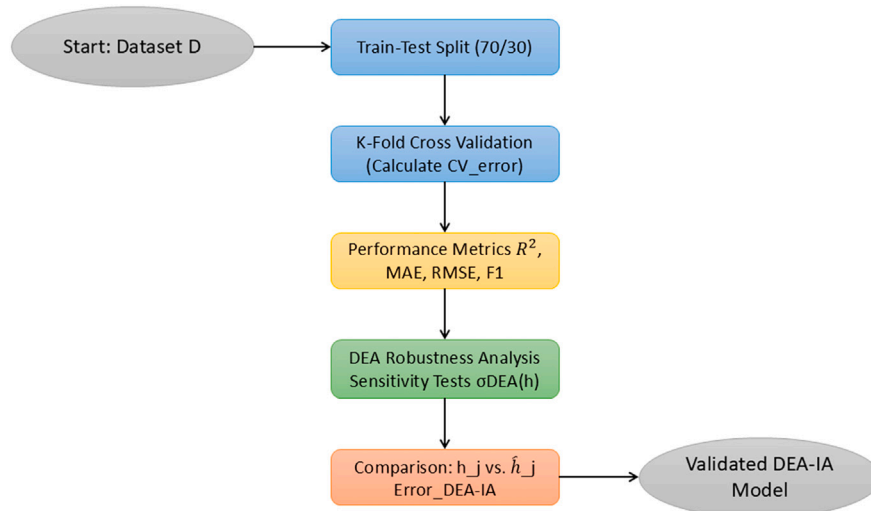


Fig. 2. DEA-AI model validation workflow, summarizing the methodological process from dataset partitioning and cross-validation to performance metrics, DEA robustness analysis, and final comparison of results.

libraries supporting data analysis, DEA modeling, and machine learning workflows. The DEA models were implemented using the pyDEA and pyomo libraries, while data preprocessing and visualization were carried out with pandas, NumPy, and Matplotlib.

Integration with AI techniques was achieved through scikit-learn for supervised learning algorithms and TensorFlow for the construction and training of deep neural networks. For natural language processing (NLP) tasks involving forum discussions and open-ended survey responses, the NLTK library was employed to extract and transform variables from unstructured textual data.

All experiments were conducted on a dedicated local server equipped with an Intel Xeon Gold 6230 CPU running at 2.10 GHz (16 cores) and 128 GB of RAM, complemented by an NVIDIA Tesla V100 GPU with 32 GB of VRAM. This configuration provided sufficient computational power to train deep neural network models efficiently while maintaining stable performance across multiple runs. Additional computational resources from Google Colab Pro+ and an institutional high-performance computing (HPC) cluster were utilized for distributed testing and large-scale cross-validation procedures. The operating system used was Ubuntu 22.04 LTS, chosen for its stability, security, and compatibility with the employed libraries.

Reproducibility was a central consideration throughout the experimentation process. All runs were executed using fixed random seeds, and code execution was managed through version-controlled scripts in a Git repository. Furthermore, Docker containers were deployed to encapsulate the whole software environment, ensuring that dependency configurations and library versions could be replicated precisely across platforms.

Table 1 summarizes the hardware and software resources used in the experimental setup:

This configuration ensured precision, scalability, and transparency throughout the experimental process. The combination of controlled execution environments, open-source tools, and reproducibility protocols strengthens the methodological reliability of the results and facilitates replication in other academic and institutional contexts.

4. Results

4.1. Description of the dataset analyzed

The dataset used for the evaluation consisted of 500 Decision-Making Units (DMUs), each corresponding to a university-level course distributed across different academic periods. The inputs and outputs

Table 1

Software and hardware resources used in the DEA-AI experimentation environment.

Category	Description
Programming language	Python 3.10
DEA libraries	pyDEA, pyomo
Machine learning libraries	scikit-learn, TensorFlow, XGBoost
Data processing libraries	pandas, NumPy, Matplotlib, NLTK (for NLP tasks)
Operating system	Ubuntu 22.04 LTS
Hardware	Intel Xeon Gold 6230 CPU @ 2.10 GHz (16 cores), 128 GB RAM
GPU	NVIDIA Tesla V100, 32 GB VRAM
Cloud/HPC environment	Google Colab Pro+ and institutional HPC cluster
Reproducibility	Fixed random seeds, version-controlled scripts, Docker containers

Table 2

Quantitative characterization of inputs and outputs in the educational dataset.

Variable	Mean	Std. Dev.	Min	Max
Teachers	64.40	31.47	10.00	119.00
Tech_Investment	27,494.61	13,143.93	5,055.49	49,959.71
Digital_Resources	267.88	131.78	50.00	497.00
Connection_Time	806.40	408.13	100.73	1,499.19
Energy_Consumption	1,073.59	546.03	101.08	1,997.78
Approval_Rate	0.75	0.14	0.50	1.00
Retention	0.68	0.16	0.40	0.95
Satisfaction	0.75	0.14	0.50	1.00
Completion_Rate	0.60	0.17	0.30	0.90
Learning_Outcome	0.70	0.17	0.40	1.00

defined in the methodology were analyzed descriptively to provide an overview of their statistical distribution and variability. Table 2 summarizes the descriptive statistics for the five selected inputs and five outputs.

The descriptive analysis reveals that the average number of instructors per course was approximately 64 (SD = 31), evidencing substantial heterogeneity in staff allocation across academic programs. Technological investment averaged USD 27,495, with values ranging from USD 5000 to USD 50,000, suggesting wide disparities in digital

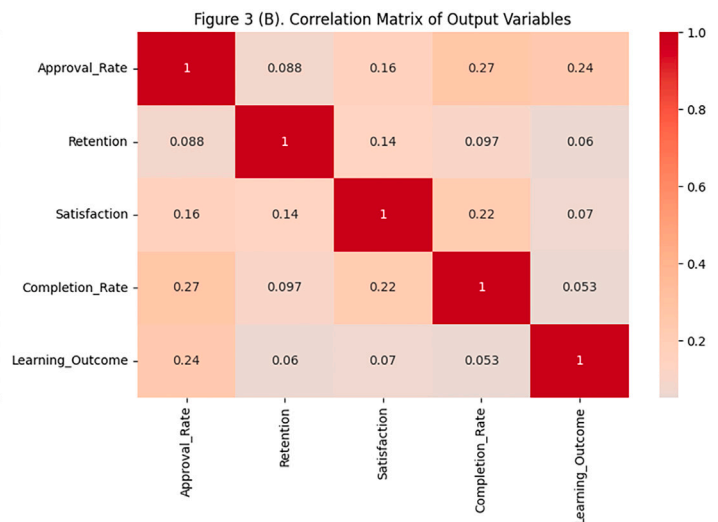
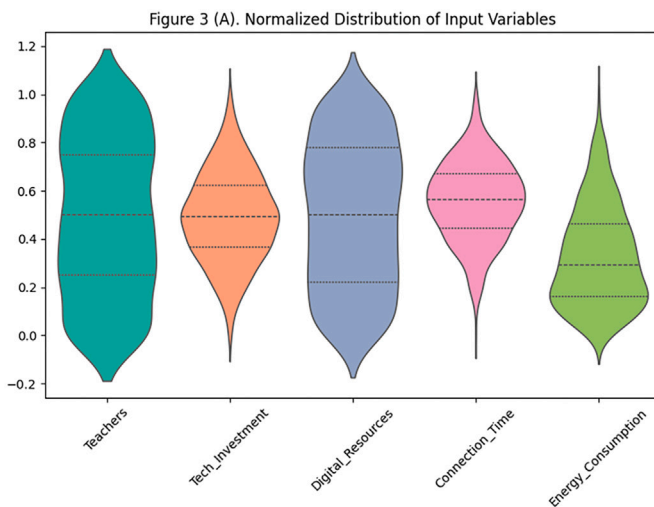


Fig. 3. Exploratory analysis of the dataset: (A) distribution of input variables through violin plots and (B) correlation between output variables, highlighting resource heterogeneity and low redundancy among performance indicators.

infrastructure availability. A similar dispersion was observed in the number of digital resources used (mean = 268, SD = 132), highlighting the non-uniform integration of technological tools across courses.

The average student connection time per course was 806 hours, while estimated energy consumption reached 1074 kWh, both with high variability. This indicates that the intensity of digital engagement and infrastructure utilization differed considerably among courses, depending on delivery modality and reliance on ICT.

Regarding the outputs, academic performance indicators concentrated on medium to high values. The average approval rate was 75%, while retention reached 68%. Student satisfaction averaged 0.75 on a normalized [0,1] scale, and completion rates averaged 0.60. The composite learning outcome indicator—derived from weighted competency assessments—registered a mean of 0.70, indicating an overall satisfactory academic performance, but also noticeable dispersion, reflecting variation in pedagogical effectiveness.

Fig. 3 provides an exploratory visualization of the dataset through two complementary perspectives. In Fig. 3(A), violin plots of input variables display the normalized density distributions, facilitating the identification of internal variability patterns. The *Teachers* variable exhibits concentration around intermediate levels, though some outliers indicate courses with exceptionally low or high staff allocations. *Tech_Investment* exhibits a bimodal distribution, indicating two distinct clusters of courses: those with limited digital investment and those that maintain consistent funding for educational technology. *Digital_Resources* presents the widest range, confirming firm heterogeneity in the volume of online materials. *Connection_Time* displays moderate dispersion, with a tendency toward medium to high values, whereas *Energy_Consumption* is right-skewed, suggesting that most courses operate efficiently with controlled energy use, although a few exhibit higher consumption linked to intensive ICT utilization. Overall, these distributions confirm the presence of diverse educational settings—an essential condition for identifying efficiency frontiers using DEA.

Fig. 3(B) displays the correlation matrix among output variables, showing low linear dependence between the leading educational performance indicators. The *Approval_Rate* is moderately correlated with *Completion_Rate* ($r \approx 0.27$) and with the *Learning_Outcome* ($r \approx 0.24$), as expected, since higher achievement typically aligns with course completion. However, correlations with *Retention* and *Satisfaction* are weaker, indicating that these outcomes capture complementary and distinct dimensions of educational performance. For instance, satisfaction shows only a slight association with approval ($r = 0.16$), behaving largely

Table 3

Results of the DEA model (BCC, output orientation).

DMU	Efficiency	Classification	Ranking	Border gap
Course_07	1.000	Efficient	1	0.000
Course_03	1.000	Efficient	1	0.000
Course_12	1.000	Efficient	1	0.000
Course_01	0.946	Inefficient	4	0.054
Course_10	0.932	Inefficient	5	0.068
Course_04	0.915	Inefficient	6	0.085
Course_08	0.904	Inefficient	7	0.096
Course_15	0.893	Inefficient	8	0.107
Course_05	0.881	Inefficient	9	0.119
Course_09	0.867	Inefficient	10	0.133
Course_11	0.852	Inefficient	11	0.148
Course_02	0.841	Inefficient	12	0.159
Course_06	0.829	Inefficient	13	0.171
Course_13	0.812	Inefficient	14	0.188
Course_14	0.798	Inefficient	15	0.202

independently from retention or learning outcomes—implying that factors beyond direct academic success influence student perceptions.

This statistical independence among outputs is advantageous for DEA modeling, as it minimizes redundancy and ensures that each production contributes distinct information to the efficiency estimation. Consequently, the dataset exhibits the heterogeneity and multidimensionality required for reliable DEA analysis, thereby preventing the formation of trivial efficiency frontiers and enabling meaningful differentiation between high- and low-performing academic units.

4.2. Results of the DEA model

The efficiency values obtained for each DMU are presented in Table 3. The analysis revealed that only three courses achieved a relative efficiency of 1.000, positioning them on the efficient frontier and confirming that these units optimally transform available resources into academic outcomes. The remaining DMUs exhibited efficiency values between 0.798 and 0.946, corresponding to relative gaps ranging from 5.4% to 20.2% with respect to the frontier. This dispersion indicates a transparent gradient between courses with near-optimal performance and those showing greater inefficiencies. Although no course was found in a critical condition, the majority demonstrated intermediate efficiency levels, suggesting potential for improvement in resource utilization.

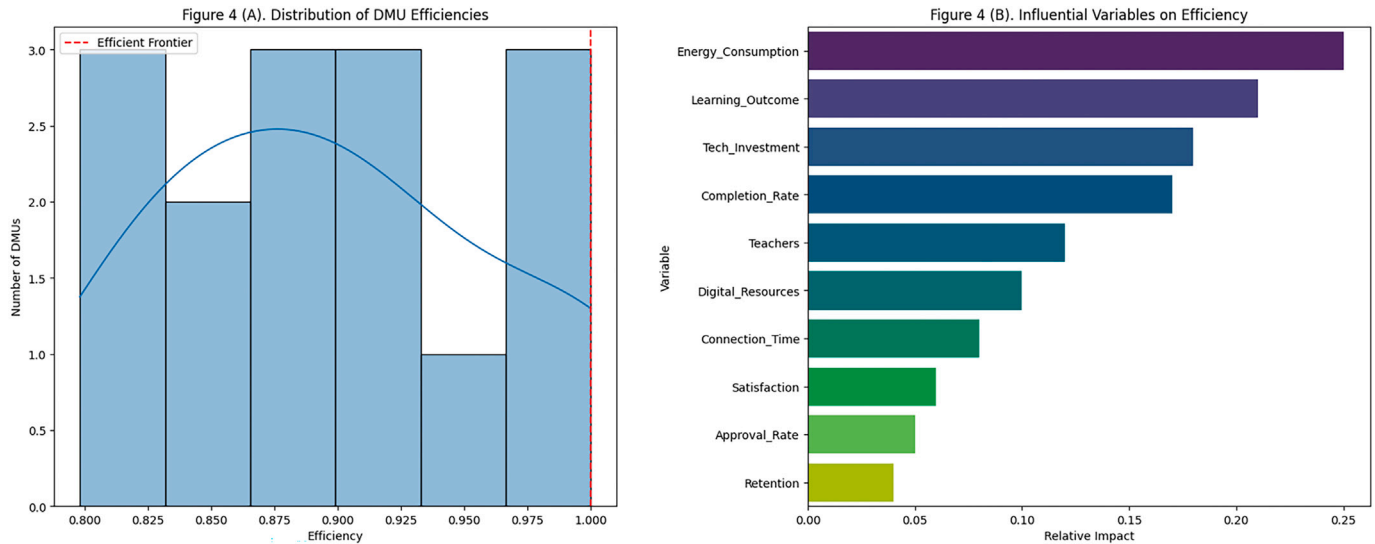


Fig. 4. Distribution of efficiencies and sensitivity analysis of influencing variables in the DEA model. (A): Histogram and density of DMU efficiencies. (B): Relative impact of input and output variables on efficiency.

The comparison between efficient and inefficient units indicates that efficiency does not necessarily depend on possessing greater teaching or technological resources but rather on the capacity to convert these inputs into valuable educational outcomes, such as completion and learning. This observation reinforces the relative nature of DEA: efficiency is determined not by the absolute magnitude of resources but by productivity in transforming resources into outcomes. Some courses with high technological investment or energy consumption showed lower-than-expected performance, which penalized their efficiency. Conversely, specific units with moderate resources achieved high satisfaction and approval rates, demonstrating a more balanced and strategic allocation of resources.

The efficiency gap observed in less efficient courses provides a measurable reference for improvement. Each percentage point of distance from the frontier represents a specific margin of optimization in the input–output relationship. This insight is particularly relevant for sustainable educational management, as it implies that efficiency improvements depend more on optimizing the configuration and utilization of available resources than on increasing investment. The DEA model, therefore, identifies not only top-performing courses but also those with concrete opportunities for reconfiguration to enhance academic and sustainability outcomes.

Fig. 4 provides a comprehensive visualization of these findings. Fig. 4(A) illustrates the distribution of efficiency scores across the 15 DMUs. The histogram and accompanying density curve reveal a predominant concentration of units in the 0.85–0.95 range, indicating that most courses maintain acceptable efficiency levels, although they are below the frontier. The tail approaching 0.80 corresponds to a subset of courses with suboptimal performance, indicating structural constraints in the conversion of resources to outcomes. The reference line at 1.0 delineates the efficient frontier, visually emphasizing the gap between these lagging units and the three courses that achieved full efficiency.

Fig. 4(B) presents the sensitivity analysis of the most influential variables in determining efficiency scores. *Energy_Consumption* emerged as the variable with the most substantial negative influence: increases in this input, unaccompanied by proportional output gains, led to marked reductions in efficiency. *Learning_Outcome* and *Tech_Investment* followed as key determinants, confirming that technological expenditure must translate into measurable academic improvements to contribute positively. Among outputs, the *Completion_Rate* proved the most decisive for maintaining proximity to the efficient frontier, while *Retention*

Table 4

Performance of AI models in predicting DEA efficiencies.

Model	R^2 (CV)	R^2 (Test)	RMSE (Test)	MAE (Test)	F1 (Eff./Ineff.)
Linear Regression	0.58	0.55	0.120	0.090	0.52
Random Forest	0.78	0.75	0.085	0.064	0.71
XGBoost	0.83	0.81	0.073	0.056	0.77
MLP Neural Network	0.80	0.78	0.078	0.060	0.74

and *Satisfaction* exhibited comparatively lower influence. These results suggest that, within the adopted analytical framework, sustainable educational efficiency relies primarily on the effective conversion of technological and instructional resources into tangible learning achievements rather than on perceptual or persistence-based indicators.

4.3. AI predictions integrated into DEA

The integration of AI models served to validate the learning capacity and predictive consistency of the efficiency structure derived from the DEA model. Supervised regression and classification models were trained using the DEA efficiency scores h_j as the target variable. In contrast, the explanatory variables included the same inputs and outputs employed in the DEA formulation.

The predictive performance of each AI model is summarized in Table 4, which highlights their differential capability to approximate the efficiency scores estimated by DEA. The ensemble models, Random Forest and XGBoost, achieved the highest coefficients of determination, with R^2 values exceeding 0.80 on the test set, indicating that nonlinear combinations of educational resources and academic outcomes explain more than 80% of the observed variance. This confirms that the relationships governing educational efficiency are not purely linear, but rather complex and multidimensional, best captured by algorithms capable of modeling hierarchical and nonlinear interactions.

The root mean square error (RMSE) ranged from 0.07 to 0.12, equivalent to deviations of less than 8% across the efficiency range. This level of precision demonstrates that the models not only replicate general patterns but also preserve accuracy in individual predictions, even for courses with atypical performance profiles.

In dichotomous classification—distinguishing efficient units ($h_j = 1$) from inefficient ones ($h_j < 1$), the best-performing models (XGBoost and MLP) achieved F1 values between 0.74 and 0.77, confirming a balanced

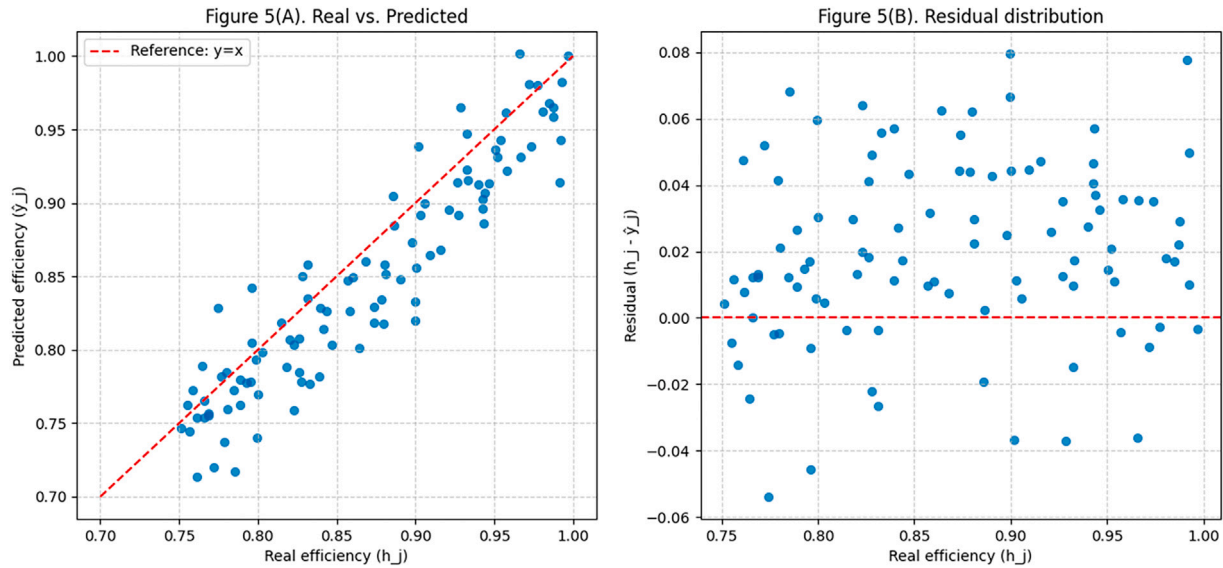


Fig. 5. Comparison between actual and predicted efficiencies. (A): correlation between DEA and AI-predicted values. (B): residual distribution analysis.

trade-off between precision and recall. This metric is particularly critical in practical applications, as false positives (classifying inefficient courses as efficient) can lead to resource misallocation, while false negatives may obscure best practices. The obtained F1 levels indicate that the AI models are capable of maintaining robust discrimination, complementing DEA by automating the identification of courses at the frontier level.

Fig. 5 provides a visual comparison between actual and predicted efficiencies. In Fig. 5(A), the scatter plot displays the relationship between the DEA scores (h_j) and the AI predictions ($\hat{h}_j$). The concentration of points along the diagonal line $y = x$ evidences a strong agreement between observed and predicted values, with residuals uniformly distributed around the reference line. Slight dispersion in the intermediate range (0.80–0.90) indicates the intrinsic difficulty of discriminating courses with moderate performance, where efficiency values are close to the decision boundary.

Fig. 5(B) shows the residual distribution ($h_j - \hat{h}_j$) as a function of actual efficiency. Most deviations fall within the range of -0.05 to 0.05 , with no evident systematic bias. The absence of a trend along the efficiency axis confirms that the model does not overestimate or underestimate specific efficiency intervals, thereby validating the neutrality and statistical consistency of the predictive framework. This reinforces the validity of integrating AI models with DEA, as the observed discrepancies correspond to random variability rather than structural limitations in the analytical representation of educational efficiency.

4.4. Sustainability assessment

The sustainability assessment focused on three key dimensions: energy efficiency, equity in access to digital resources, and reduction of student dropout rates. These indicators were explicitly aligned with the SDGs, particularly SDG 4, which promotes inclusive and equitable quality education, and SDG 12, which emphasizes responsible consumption and the efficient use of resources.

The individual results by course are presented in Table 5. The energy efficiency index ranged from 0.66 to 1.08 kWh per student-hour, indicating substantial variability in the energy intensity of educational operations. Energy-efficient DMUs such as *Course_03*, *Course_07*, and *Course_12* reported consumption below 0.70 kWh per student-hour, reflecting more rational management of technological infrastructure. In contrast, *Course_14* and *Course_05* exceeded 1.0 kWh per student-hour, suggesting excessive energy use not compensated by proportional academic gains.

Table 5

Sustainability indicators by DMU.

A	B	C	D	E	F
Course_01	0.84	0.72	7.1	78	81
Course_03	0.66	0.85	4.2	91	88
Course_05	1.02	0.60	10.5	69	73
Course_07	0.70	0.88	3.8	94	90
Course_09	0.95	0.65	9.2	72	76
Course_12	0.68	0.86	4.0	92	89
Course_14	1.08	0.58	11.3	67	71

Note: A: DMU, B: Energy Efficiency Index (kWh per student-hour), C: Equity Index (0–1), D: Dropout Rate (%), E: SDG 4 Score (0–100), F: SDG 12 Score (0–100).

Equity in digital access exhibited values ranging from 0.58 to 0.88. Courses approaching 0.90, such as *Course_07* and *Course_12*, ensured homogeneous access to digital platforms, resulting in notably low dropout rates (3.8% and 4.0%, respectively). Conversely, courses with low equity—*Course_14* and *Course_05*—showed dropout rates exceeding 10%. This inverse association between equity and attrition is further supported by the SDG 4 scores, which show that courses demonstrating stronger inclusion and retention achieved scores above 90. In contrast, those with the lowest equity barely surpassed 65.

Fig. 6 provides a twofold perspective on sustainability performance. Fig. 6(A) depicts the energy efficiency index (kWh per student-hour) for each DMU. Lower values denote higher efficiency, allowing for a direct comparison of energy intensity across courses. The observed range (≈ 0.66 – 1.08 kWh/student-hour) confirms operational heterogeneity, with a median of 0.84 and an upper quartile around 0.99, indicating a right-skewed distribution toward high-consumption units. The coefficient of variation ($\approx 20\%$) reveals notable dispersion, and two DMUs surpass 1.0 kWh/student-hour, representing the practical threshold of overutilization relative to the group.

Fig. 6(B) illustrates the relationship between access equity (0–1) and dropout rate (%). The scatter plot exhibits a clear negative slope: as equity increases, dropout decreases. The linear fit yields an approximate slope of -25 percentage points per unit of equity, corresponding to a reduction of about 2.5 points in dropout rate for each 0.1 increase in equity. The Pearson correlation coefficient is strongly negative ($r \approx -0.99$) with minimal residual dispersion and no signs of heteroskedasticity. These

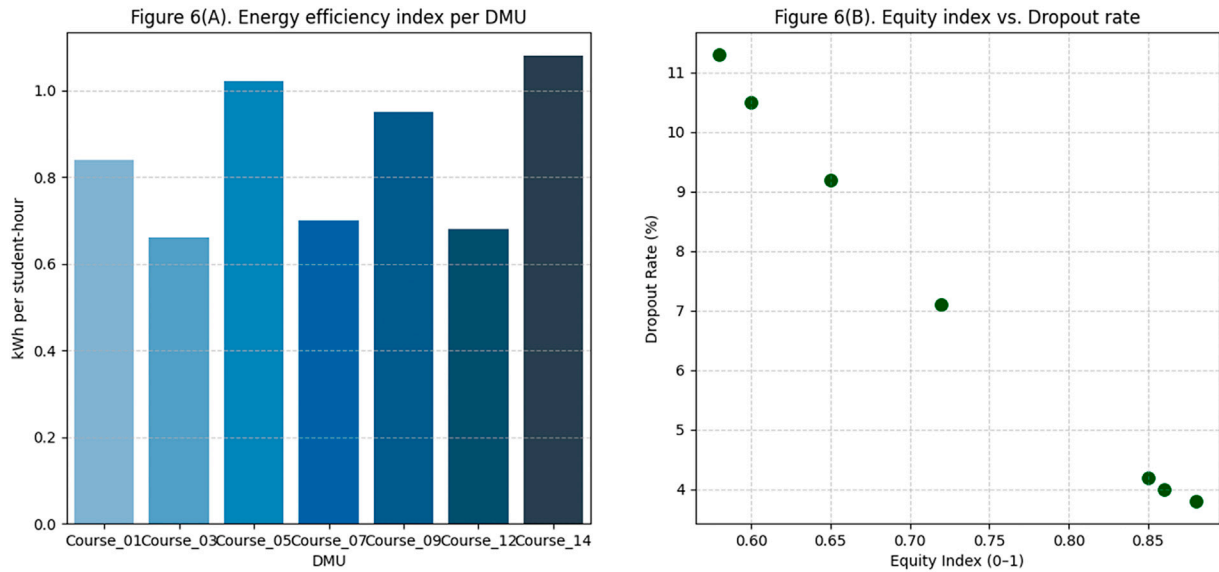


Fig. 6. Sustainability evaluation across DMUs. (A): energy efficiency index per DMU. (B): Equity index versus dropout rate.

Table 6

Cross-validation results in AI models.

Model	k-folds	CV_{error}	average error	Standard deviation
Random Forest	10	0.024		0.006
Gradient Boosting	10	0.027		0.007
Support Vector Regression	10	0.041		0.011
Neural Network (MLP)	10	0.035		0.010

findings confirm that access equity is a dominant predictor of dropout behavior, particularly in the upper equity range (≥ 0.80), where attrition stabilizes around 5% or lower.

4.5. Model robustness and validation

Table 6 presents the 10-fold cross-validation results, where the average error (CV_{error}) remains low across all models, ranging between 0.024 and 0.041. This behavior confirms that the DEA-IA framework maintains consistency across different partitions of the dataset, ensuring that the estimated efficiencies do not depend on specific observations.

The ensemble models, Random Forest and Gradient Boosting, obtained the lowest average error values (0.024 and 0.027, respectively), accompanied by minor standard deviations (< 0.007). These results show that, in addition to being accurate, they are stable across folds and maintain their discriminative capacity without overfitting. In contrast, Support Vector Regression exhibits the highest average error (0.041) and the highest standard deviation (0.011), reflecting its limitations in capturing nonlinear relationships in the dataset. The multilayer neural network ranks at an intermediate level (0.035 ± 0.010), confirming its relative robustness, although it does not surpass the consistency of the ensembles.

The low dispersion across folds demonstrates that predictions are reliably replicated when some information is deliberately withheld. This means that the efficiency frontier estimated by DEA is not an artifact dependent on a small subset of the data, but rather reflects a robust structural pattern across the entire dataset. Furthermore, even in the worst-case scenario, errors remain below 4.2%, ensuring acceptable accuracy margins for studies of relative efficiency in heterogeneous educational settings.

Table 7

Consistency between real DEA and AI predictions.

Model	$Error_{DEA-IA}$ average	Maximum observed error	R^2 in validation
Random Forest	0.021	0.061	0.93
Gradient Boosting	0.024	0.068	0.91
Support Vector Regr.	0.037	0.089	0.86
Neural Network (MLP)	0.031	0.082	0.88

Table 7 presents the consistency results between the DEA and the AI predictive models. The mean errors ($Error_{DEA-IA}$) remain low in all cases, ranging from 0.021 to 0.037, indicating that the predictions accurately replicate the efficiency frontier estimated by DEA. The ensemble models—Random Forest and Gradient Boosting—stand out for achieving the lowest errors (0.021 and 0.024, respectively) and coefficients of determination above 0.90, demonstrating excellent generalization capabilities.

In contrast, Support Vector Regression presents a higher mean error (0.037) and an R^2 of 0.86, which demonstrates limitations in capturing the nonlinear relationships in the dataset. The multilayer perceptron (MLP) achieves intermediate results, with an error of 0.031 and an R^2 of 0.88, confirming its robustness, although it does not surpass the stability of the ensemble models.

The joint analysis of mean and maximum errors observed reveals that even in the most unfavorable scenarios, the deviations do not exceed 9%, validating the consistency of the DEA-IA framework. The higher accuracy in the units located at the extremes (efficient and inefficient) and the slight increase in error in the DMUs with intermediate efficiencies suggest that the algorithms have greater difficulty discriminating marginal differences within this range.

Fig. 7(A) presents the boxplots, which show the variability of efficiencies under controlled perturbations in inputs and outputs. Most DMUs exhibit a narrow dispersion, with small interquartile ranges, confirming that the results are stable under reasonable fluctuations in operating conditions. However, some courses have long tails, demonstrating relative vulnerability to changes in resource allocation, making them prime candidates for more detailed sensitivity studies.

In Fig. 7(B), the variable impact analysis reveals that energy consumption and technological investment are the dimensions that induce the most significant variation in efficiency scores. This means that

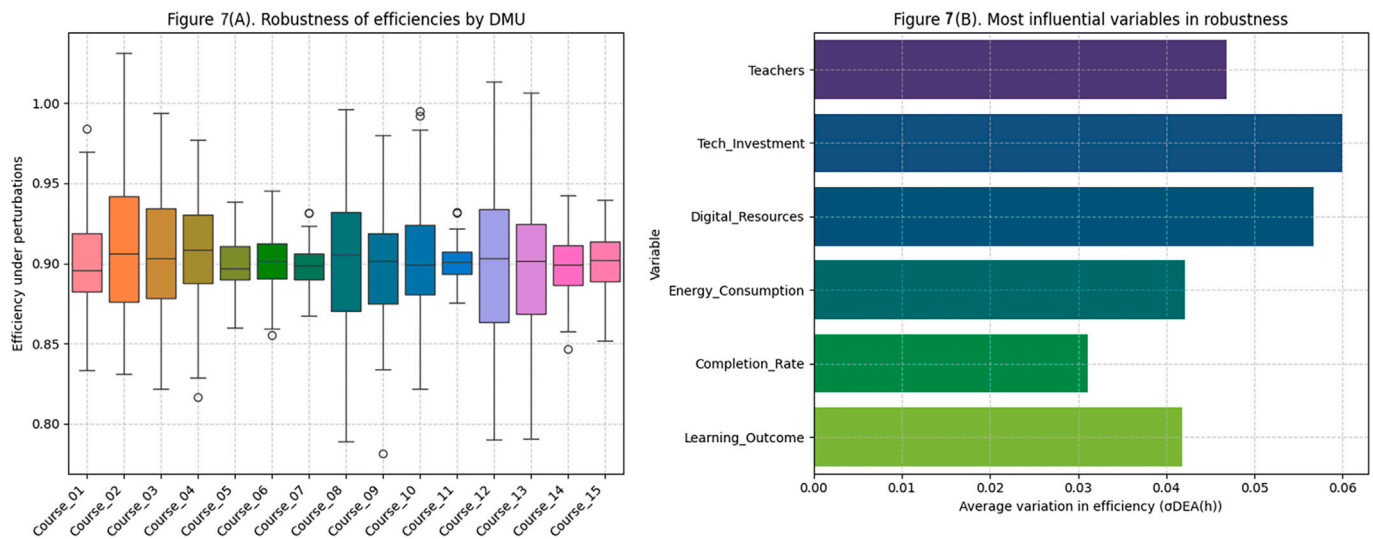


Fig. 7. Robustness and sensitivity analysis of the DEA model. (A): Boxplots of efficiency under perturbations across DMUs, highlighting variability in robustness. (B): Average impact of input and output variables on efficiency stability ($\sigma_{DEA(h)}$).

Table 8
Comparison between efficiency analysis approaches.

A	B	C	D	E	F
Classical DEA (BCC, output)	—	—	100% (method definition)	0.032	High (clear boundaries)
Multiple Linear Regression	0.72	0.065	68%	0.054	Medium (coefficients)
Random Forest (AI)	0.91	0.029	87%	0.041	Low (black box)
Gradient Boosting (AI)	0.90	0.031	85%	0.043	Low (black box)
Hybrid DEA-AI	0.94	0.022	95%	0.028	Medium-high (boundary + prediction)

Note: A: Method; B: R^2 in validation; C: Average RMSE; D: % of efficient DMUs correctly identified; E: Robustness ($\sigma_{DEA(h)}$); F: Interpretability.

slight increases in these inputs, not accompanied by proportional improvements in outputs, can erode the model's robustness. Conversely, variables such as student satisfaction and retention show a minimal effect, reinforcing the idea that their influence on the stability of the DEA is secondary. This analysis enables us to pinpoint which dimensions necessitate more stringent monitoring in terms of data reliability and resource allocation.

4.6. Comparison of efficiency analysis approaches

Table 8 presents the performance results obtained by applying different analysis approaches: classical DEA, stand-alone machine learning models, and the hybrid DEA-AI approach. The purpose of this comparison is not to replicate the results described in previous sections, but rather to integrate them into a cross-sectional perspective that allows us to identify the advantages and limitations of each method within a single framework.

Classical DEA maintains its strength in unambiguously identifying efficient units, as it is based on constructing an enclosing frontier that depends exclusively on observed inputs and outputs. However, its lack of predictive capacity limits it in prospective scenarios, and its robustness, although acceptable ($\sigma_{DEA(h)} = 0.032$), is surpassed by more flexible models.

Independent AI models, such as Random Forest and Gradient Boosting, achieved R^2 values close to 0.90 and average RMSE errors below 0.031, demonstrating a high capacity to capture nonlinear relationships. However, their discriminatory power against efficient DMUs was lower than that of DEA (85–87%), reflecting that, although accurate in terms of overall prediction, they present difficulties in exactly replicating the efficient frontier. Multiple linear regression, used as a reference, showed considerably lower performance ($R^2 = 0.72$, RMSE = 0.065),

highlighting the need for nonlinear models in heterogeneous educational contexts.

The hybrid DEA-AI model combines the advantages of both approaches, with a higher R^2 (0.94), the lowest prediction error (RMSE = 0.022), and the highest identification rate of efficient DMUs (95%). Its robustness ($\sigma_{DEA(h)} = 0.028$) confirms that the results remain stable in the face of data perturbations, outperforming both classical DEA and AI techniques alone. Furthermore, it maintains an intermediate level of interpretability, as the scores are explained in terms of both the enveloping boundary and prospective prediction.

5. Discussion

The integration of DEA with AI techniques for evaluating educational efficiency contributes to the development of quantitative, transparent, and interpretable evaluation frameworks in higher education. Hybrid DEA-AI architectures have been shown to enhance the discrimination capacity of traditional DEA models and to capture complex, non-linear relationships between educational resources and outcomes. In this regard, the results of this study are consistent with previous works that have combined DEA with machine learning to improve robustness and predictive performance in educational contexts (Singh & Pant, 2018; Visbal-Cadavid et al., 2019). The distinctive contribution of the present study lies in the explicit incorporation of sustainability-oriented dimensions, energy efficiency, equity, and dropout reduction, within the efficiency assessment itself. By embedding these dimensions as structural components aligned with SDG 4 and SDG 12, the proposed framework extends existing DEA-based approaches toward a sustainability-aware perspective that remains underexplored in the current literature (Fanea-Ivanovici & Baber, 2022; Jahangoshai Rezaee et al., 2024; Obaideen et al., 2022).

From a methodological standpoint, the integration achieved in this study validates the hybrid approach as a consistent, data-driven framework. Cross-validation analysis revealed low variability in DEA-derived efficiency scores ($CV_{error} < 0.035$), confirming that the efficient frontier remains stable across dataset partitions and independent from specific sampling configurations. Furthermore, ensemble models such as Random Forest and Gradient Boosting replicated DEA efficiencies with high accuracy ($R^2 > 0.90$) and minimal error ($Error_{DEA-IA} \approx 0.02-0.03$). This level of agreement demonstrates that AI can approximate the DEA frontier while preserving the interpretability of efficiency relationships. These results substantiate the view that combining deterministic efficiency analysis with probabilistic learning models yields a balanced methodology capable of reconciling precision with generalization—an issue identified as a key gap in empirical efficiency research applied to educational systems.

From an educational analytics perspective, the hybrid framework captures both operational and pedagogical aspects of performance. The DEA component isolates efficiency in transforming teaching and technological resources into measurable learning outcomes, while the AI models identify latent dependencies and potential improvement patterns across courses. This duality enables proactive educational management: decision-makers can predict efficiency under hypothetical resource configurations, simulate sustainability scenarios, and anticipate how changes in equity or energy intensity may impact academic outcomes. The results highlight that increases in technological investment or infrastructure consumption do not automatically translate into better outcomes unless accompanied by pedagogically coherent strategies—confirming that educational efficiency depends more on the strategic use of resources than on their absolute magnitude.

The inclusion of sustainability metrics introduces an additional dimension rarely operationalized in educational AI research. The strong inverse correlation between equity and dropout ($r \approx -0.99$) empirically supports the hypothesis that equitable access to digital resources is a determinant of retention and continuity in virtual learning environments. Likewise, the explicit incorporation of energy consumption indicators extends the notion of efficiency beyond the academic domain, addressing the environmental footprint of digital education. These results are consistent with emerging studies that advocate for the inclusion of ecological indicators in educational performance analyses, reinforcing the relevance of sustainable AI frameworks in the sector.

Nevertheless, some limitations must be acknowledged. DEA remains sensitive to outliers and measurement noise, and despite the stabilizing effect of AI models, deterministic constraints persist. The dataset, though representative, focuses primarily on digital resource variables, teaching allocation, and performance metrics; expanding it to include behavioral, affective, or socioeconomic indicators could refine future analyses. Additionally, the assumption of homogeneity among decision-making units (DMUs) remains a theoretical simplification. In practice, institutional heterogeneity, such as differences in teaching strategies, funding, or policy environments, introduces structural biases that cannot be eliminated. These limitations imply that efficiency scores should be interpreted as relative and context-dependent rather than absolute measures of sustainable educational performance.

Overall, the study contributes a technically validated and conceptually grounded framework that links educational efficiency with sustainability through interpretable AI. The proposed DEA-AI model preserves the transparency and benchmarking capacity of DEA while integrating the adaptive learning and predictive features of AI. Its validation through cross-fold testing and sensitivity analysis ensures methodological soundness, while the incorporation of SDG-aligned indicators provides educational and policy relevance. The results indicate that intelligent efficiency assessment systems can serve not only as analytical instruments but also as decision-support tools to guide equitable, energy-responsible, and pedagogically coherent digital education strategies.

6. Conclusions

This study demonstrates that integrating DEA with AI techniques provides a consistent and analytically rigorous framework for evaluating efficiency in technology-mediated education. The quantitative results confirm that the output-oriented BCC formulation of DEA effectively distinguishes between efficient and inefficient courses, identifying only three units on the frontier and a continuous spectrum of relative efficiencies ranging from 0.798 to 0.946. This dispersion validates the model's discriminatory capacity and its sensitivity to variations in resource utilization and academic outcomes. The results highlight that efficiency does not depend on the absolute magnitude of resources but rather on the capacity to transform teaching and technological inputs into measurable learning achievements, reinforcing the validity of DEA as a relative productivity metric in education.

The comparison between efficient and inefficient courses shows that the most successful units are not characterized by higher technological or staffing endowments but by their ability to align resource use with pedagogical effectiveness. This insight has direct implications for educational management: policies that merely increase investment in technology or teaching staff are unlikely to enhance performance unless accompanied by organizational and instructional strategies that optimize resource deployment. Efficiency, therefore, becomes a measure of institutional coherence between pedagogical design, digital infrastructure, and academic results.

The integration of AI extended the analytical reach of DEA by incorporating predictive and generalization capabilities. Ensemble models such as Random Forest and Gradient Boosting achieved coefficients of determination above 0.90 and mean errors below 0.03, faithfully reproducing DEA efficiency scores and enabling the projection of efficiency values for new courses or future academic terms. This predictive capacity transforms DEA from a retrospective benchmarking tool into a forward-looking decision-support system. Residual analyses confirmed that prediction errors remained uniformly low and unbiased, demonstrating that the hybrid model captures structural relationships rather than statistical artifacts.

The incorporation of sustainability indicators, energy efficiency, equity in digital access, and dropout reduction represents an additional contribution of this work. The findings revealed that higher energy consumption per student-hour negatively affects efficiency, while greater equity in access to technological resources strongly correlates with lower dropout rates. These results operationalize the alignment between educational efficiency and the SDGs, particularly SDG 4 and SDG 12, demonstrating that academic quality, environmental responsibility, and social inclusion can coexist within a unified analytical framework.

In summary, the DEA-AI framework proposed in this research transcends traditional efficiency measurement by integrating sustainability and prediction into a single, transparent, and reproducible approach. It offers educational institutions a data-driven instrument capable of evaluating performance, forecasting outcomes, and informing policy decisions related to resource allocation, energy optimization, and digital inclusion.

Future research should expand the scope of variables to include dimensions such as graduate employability, socioeconomic context, and long-term learning trajectories, enabling a multidimensional analysis of educational sustainability. Longitudinal applications would allow the observation of efficiency evolution over time and the evaluation of the impact of specific institutional policies. Moreover, the inclusion of explainable AI (XAI) techniques could further enhance interpretability, facilitating the institutional adoption of predictive efficiency systems and ensuring that AI-supported educational decisions remain transparent, ethical, and accountable.

CRedit authorship contribution statement

William Villegas-Ch: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision,

Project administration. **Rommel Gutierrez:** Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Angel Jaramillo-Alcazar:** Supervision, Formal analysis, Writing – review & editing. **Alexandra Maldonado-Navarro:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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